

Subject Field: Natural Science Subjects

1. General Principles

1.1. Field-related competencies

The purpose of natural science subjects in basic school is to develop students' competence in natural sciences, as a result of which the students:

- 1) take an interest in the environment and its study and the study of natural sciences;
- 2) apply the knowledge and skills acquired in natural science subjects to explain and analyse environmental objects, phenomena and cause-and-effect relationships between them, using natural science terminology and natural science models;
- 3) notice, formulate and solve problems related to everyday life, make reasoned decisions and use creative and critical thinking;
- 4) formulate research questions related to the natural sciences, design and conduct a study following safety requirements, and draw evidence-based conclusions;
- 5) find information on natural sciences and technology from a variety of sources and assess its reliability; use media and technology tools for learning, data collection and collaboration;
- 6) understand the nature, importance and limitations of science, the links between science and technology, and the risks;
- 7) value biodiversity and sustainable development, behave safely and lead a healthy lifestyle;
- 8) know career opportunities related to natural sciences and technology and are motivated for lifelong learning.

1.2. Estimated volume of subjects in the subject field

The subject field includes five subjects, of which natural science is studied from the 1st to 7th grade; biology and geography from the 7th grade; and physics and chemistry from the 8th grade. The estimated lessons per week per school stage to achieve the learning outcomes described in the syllabuses are as follows:

subject	First school stage	Second school stage	Third school stage
natural science	3	7	2
biology			5
geography			5
physics			4
chemistry			4

The distribution of lessons per week within the school stages and the teaching content by grade is determined in the school curriculum, taking into account the learning outcomes and the knowledge, skills and attitudes to be acquired by the end of the school stages.

1.3. Description of the Subject Field and Integration within the Subject Field

Competence in natural science, which includes knowledge of natural sciences, research and problem-solving skills and attitudes that value sustainable development, is essential for everyone in today's world. It helps to notice the problems of everyday life and make smart and reasoned decisions using knowledge and skills related to natural science. In addition to coping with personal life, competence in natural science supports self-realisation at work, because the labour market increasingly needs creative, critically thinking people who work in the field of natural science and technology and constantly improve their knowledge and skills.

The core of scientific competence is a worldview based on natural science, a scientific way of thinking and an attitude that values it, characterised by curiosity about the surrounding phenomena, open but critical thinking and a consistent pursuit of evidence-based and unbiased knowledge.

A conceptual understanding of the subject is formed only when new knowledge is linked to existing knowledge and experiences and knowledge gained in other natural science subjects. In addition to describing the directly perceived world, students gradually learn to describe objects and phenomena at the micro and mega levels and use natural science symbols. It is important to develop an understanding of the cause-and-effect relationships of phenomena and to generalise and transfer the learning to a new context. A variety of natural science models contribute to generalisation, including physical objects, drawings, maps, concept maps, mathematical representations, analogies and computer simulations. Models help to understand, study and explain natural objects and phenomena and to make conclusions and predictions on the behaviour of objects and systems. Students build models themselves, with the teacher expected to help students understand the limitations of the models. Studying subjects in the field of natural sciences helps students to understand the nature of science and scientific knowledge. This means, in particular, that scientific knowledge is evidence-based and refutable if more accurate and compelling research results emerge — these circumstances distinguish scientific knowledge from personal, religious, political or other beliefs. Students should understand that scientific views change over time and evolve toward an increasingly accurate and objective description of the world. It is important to understand the limitations of science, which means that the conclusions drawn only apply to organised research. Taking the results out of context, ie over-generalising or simplifying, can lead to invalid conclusions.

All natural science subjects develop students' research skills that include observing objects and phenomena, defining problems, collecting and analysing background information, formulating research questions and hypotheses, designing and organising experiments, collecting reliable data, analysing and interpreting it and drawing valid conclusions. The general goal of acquiring research skills is to use them in everyday life, helping students make intelligent and deliberate decisions in their personal lives.

Students' communication skills develop as they study natural subjects. In the information society, the skills of searching for, understanding and interpreting information related to natural sciences are increasingly important. With the increasing use of social media and alternative sources of information, students need help in distinguishing reliable and evidence-based information from someone's personal opinion. Students' self-expression skills develop through the presentation of research results, project work, etc, and through compiling written text. They will also develop their ability to discuss problems and justify their proposed solutions based on scientific, social, economic, ethical and other perspectives.

In natural science classes, giving meaning to values plays an important role, that is, discussing and justifying them based on the student's own point of view as well as that of others, and learning to take into account different points of view. It is important to develop an understanding that society can only develop sustainably if everyone contributes to the preservation of biodiversity and a habitable living environment.

In order for students to wish to continue their studies in natural sciences and technology, they need an understanding of the diversity and specificity of these specialities. Already by basic school, students should be helped to set personal subject-related goals to enable further learning at the next school stage and to make primary career choices.

Through the integration of natural science subjects, students develop an understanding of the natural and artificial environment as an integral system and the parts of each natural science subject in this whole. Natural science subjects are integrated at three levels: through the development of competence in natural science, overlapping teaching content, ie thematic integration, and co-operation between the school curriculum and natural science teachers.

1.4. Opportunities for cross-disciplinary integration to support the formation of general competencies and to address leading topics of the curriculum

Natural science subjects support the formation of all general competencies described in the general part of the national curriculum.

The achievement of general competencies is supported by the cross-disciplinary integration of subjects with subjects in other fields and by the meaningful handling of leading topics. As a result, students will develop the ability to apply their knowledge and skills in different situations, shape their values and attitudes and gain an understanding of the development of society as a whole. Systematic and consistent co-operation between subject teachers is essential.

The formation of general competencies and the principles for addressing leading topics and organising integration are defined in the general part of the school curriculum and implementation is specified in the subject field syllabus.

1.5. Planning and organising learning activities

The planning and organisation of studies are based on the fundamental values of the curriculum, general competencies, the priorities of teaching and education at school stages, the competencies and learning outcomes in natural sciences and the teaching content provided for in the school curriculum. In addition, integration with other subjects and leading topics is supported.

In teaching and learning natural sciences, it is vital that students are able to take independent action and experience the thrill of discovery that accompanies understanding the world around them and testing their abilities. Experiencing and giving meaning to it help to develop deep foundational knowledge, but also skills and attitudes that support the student's ability to cope in everyday life and, more broadly, the functioning of a democratic and sustainable society. This approach to learning is supported by a variety of teaching methods: exploratory work, including practical work, discussions, research and construction of natural science models, debates, project study, role play games, presentations, mutual teaching, etc.

The content of the subject is learned, skills are developed and attitudes are shaped in a problem-based manner close to everyday life, which helps to make sense of the learning and its necessity. Decision-making, troubleshooting, problems related to design or dilemmas, etc should be personally and/or socially relevant to the student.

Learning and teaching is based on an exploratory approach that takes into account students' interests and questions and supports their self-initiative. Assignments take into account students' abilities, are age-appropriate and support their development. Learning difficulties are responded to and learning assistance will be provided where necessary. Particular attention should be paid to the specific characteristics of the student, including talent in the subject.

Students' study load, including the volume of homework, is distributed evenly throughout the school year and leaves sufficient time for rest and hobbies;

Group and pair work develop students' co-operation and planning skills, their consideration of different points of view and consideration for others as well as tolerance of criticism. Presenting works and interacting with each other develop the students' self-expression skills.

In the course of studying natural sciences, the student develops an awareness of specialities and professions related to natural sciences and technology, which is introduced in everyday studies and by inviting guest speakers and visiting institutions. An overview of the content of the work, the working conditions, the required skills and the educational prerequisites allow the student to consider the suitability of their interests and abilities to a particular speciality.

Diverse learning methods, problem-based and exploratory approaches, collaborative learning and the use of modern learning environments help to increase students' learning motivation and to form independent learners.

1.6. Assessment

Assessment is part of the learning process, supporting student learning and development. Assessment provides an overview of the achievement of learning outcomes and the individual development of the student and supports the development of the student as a self-guided learner with a positive self-image and sufficient self-esteem. As a result of the assessment, students receive feedback on their learning progress and for choosing learning strategies. Teachers will receive information on their teaching performance and input for the development of both their teaching and their own competencies.

Assessment is based on the learning outcomes described in the syllabus of the subject field by school stages. Assessment supports the acquisition of the knowledge and skills to be acquired by the end of the school stage, the development of attitudes and the achievement of competencies in the related field. Alongside subject knowledge and skills, feedback is also provided on the formation of general competencies and the development of values. Feedback on the formation of attitudes is provided through guiding and supportive verbal assessments.

Students receive formative assessment during the course of study and summative assessment at the end of the topics and the school stage. Assessment should be consistent with general competencies, competencies in the related field and the learning outcomes sought. Thus, it should be varied to support the development of the student's knowledge and different skills and attitudes.

With diagnostic assessment, the teacher identifies the student's strengths and weaknesses, including misconceptions related to natural sciences and specific learning difficulties, at the beginning of the academic year or topic, to plan future teaching.

During teaching, the teacher provides feedback on the student's performance so that the student can immediately know how their learning is progressing. Written feedback is provided by the teacher on an ongoing basis for the further development of large-scale work, such as a portfolio, essay, research paper, etc. Students receive automatic feedback on tasks solved in online environments which allows them to evaluate their knowledge and skills, set goals and improve results.

Assessment is used as part of learning when students evaluate work completed by themselves or their peers based on agreed criteria. This way, students learn to notice their mistakes and analyse them.

Students discuss learning independently, in a group or with a teacher – what were the positive aspects of the work and what could be done better next time.

Assessment criteria, or assessment models, are particularly needed for the successful performance of learning tasks that require open and/or creative thinking (research, presentations, essays, poster presentations, science models prepared by students, etc). Assessment models make the students more aware of the teacher's expectations, allow them to manage their learning and provide constructive feedback to their peers. In addition, they help the teacher to assign the final grade when the work is completed, and the student to better understand how the grade was formed.

Both formative and summative assessments focus primarily on understanding what has been learned, on discussing and using knowledge, rather than merely recalling what has been learned. The fact that as the students become older, the significance of more complex and demanding knowledge and skills increases, is taken into account in the choice of assessment methods/forms.

In addition to tests and test papers, the assessment includes presentations, poster presentations, reports on research, essays, natural science models created, including concept maps, collections, videos, portfolios, designs or solutions developed in the course of a project, etc.

Research skills are also assessed as subskills, which include posing a research question or designing an experiment based on a given situation or test tools, drawing conclusions based on given data, critically evaluating the quality of an organised experiment, making

suggestions to increase the reliability of test results and obtaining valid conclusions.

The student's self-esteem plays an important role in the formation of attitudes and values. The student's attitudes and values are not directly the subject of summative assessment. They are assessed through the student's ability to give meaning to values, that is, to discuss, reason and justify them based on their personal or others' perspectives.

In the case of problem-solving tasks, the assessment criteria are the feasibility of the proposed solution, the number of justifications, highlighting social, ethical, economic and other aspects, originality, logic and the correct use of natural science terminology. In the case of a natural science essay, the assessment criteria are the clarity of the posed problem, the number and logic of examples and justifications, the use of correct natural science terms, the validity of the conclusions, the overall coherence between parts of the text and the originality of the author's thoughts.

1.7. Learning environment

The school provides an inspiring, cooperation-oriented and safe learning environment where all students can experience success and receive recognition for their work and effort. The latter does not exclude being demanding and setting clear goals, provided that they are based on the actual abilities of the student. A friendly and supportive environment free of bullying and violence creates conditions for students to be able to devote themselves to learning and overcoming any difficulties encountered. Values inherent in a democratic society need to be formed. Dissenting opinions are accepted, discussed and assessed based on science-based facts and the basic principles of a democratic society. Students are involved in the planning and assessment of studies.

The learning takes place in as diverse environments as possible, including school surroundings, nature, museums, nature schools, science education centres, companies, etc. Opportunities offered by higher education institutions, such as laboratories, courses, etc, are used. State-of-the-art teaching materials and digital tools as well as e-learning environments are used in studies, which also support the development of students' digital competence.

Practical work requires test tools and materials and conditions for their storage, as well as special desks in the classroom and the ability for students to use hot water, sinks and electrical outlets. The teacher needs demonstration tools as well as technical possibilities for their use.

For practical work, large classes are divided into smaller groups where necessary. The safety and efficiency of the organisation of laboratory work is ensured.

2. Syllabuses

2.1. Natural science

2.1.1. Description of the subject

The subject aims to develop a caring attitude towards nature and the living environment and everything alive, an understanding of nature and the artificial environment (hereinafter the *environment*), and the principles of sustainable development in students. At the same time, the foundation for the formation of the student's natural scientific worldview and way of thinking is laid. The latter are characterised by curiosity about surrounding phenomena, open but critical thinking and striving for evidence-based knowledge.

Studying natural science allows students to begin to acquire an understanding of the environment as a whole. The main objects of cognition in learning are objects and phenomena found in the environment and the relationships between them. They will learn to understand the regularities of the functioning of natural phenomena and the interaction between humans and the environment. Natural science develops an understanding that every phenomenon has a cause and any change in the environment induces other changes that may be desired or undesired.

The goal of natural science is to create a solid foundation for the development of science literacy, on which other natural sciences (biology, geography, physics, chemistry) can later

rely, consisting of the following:

- 1) the ability to notice, observe and explain objects and phenomena occurring in the environment and the relationships between them; the ability to apply scientific knowledge and skills in solving problems of everyday life;
- 2) research skills: ability to formulate research questions or hypotheses that can be tested; plan experiments to collect data; perform practical work using test tools, equipment and measuring instruments in a safe manner; analyse data and their reliability; draw valid conclusions, formulate generalisations and present results;
- 3) ability to find information about the natural sciences from various sources, interpret it and evaluate the reliability of the information, use terms, units and symbols related to natural sciences in both oral and written self-expression, including when presenting information, discussing problems and reasoning their claims;
- 4) attitudes and values that support the handling of natural science issues: self-efficacy in the study of natural sciences; interest in studying natural sciences and careers in natural science and technology; preparedness to deal with issues related to natural sciences and responsibility for sustainable development.

The organisation of studies is based on experiencing the environment and age-appropriate activities. Practical activities, in the course of which objects and phenomena are studied directly, but also with the support of natural scientific models, are highlighted. Learning should support students in posing problems and questions and finding answers and solutions to them. They should be open and related to everyday life as much as possible, that is, allow different solutions. The latter also promotes the development of creative and critical thinking in students. An active, student-centred and problem-based learning environment fosters internal motivation and self-regulation in students.

In the first school stage, the students mainly get to know their surroundings and everyday phenomena, focusing on the direct experience of the environment and practical activities. By the end of the school stage, students can make easier connections and draw conclusions after learning to describe objects and phenomena. The student's interest in nature, ability to behave in nature and their environmental attitudes are formed. The primary basis for the formation of the student's natural science way of thinking is created: in the course of practical activities, the student is directed to ask simple questions and make assumptions about the surrounding substances, materials, objects and phenomena, observe, compare and group them, measure, conduct experiments, draw up collections and use a map. The student is encouraged to talk about their observations and discoveries.

II The second school stage further develops the students' natural scientific thinking and research skills. The ability to formulate smaller scientific questions and hypotheses that can be tested by experiments, to plan and carry out an experiment and draw conclusions are developed. To find answers to questions, students are encouraged to use secondary sources: popular science magazines, news portals and books, Wikipedia in Estonian or another language, etc. A primary understanding of how to find reliable information is formed. It is important to plan research-based tasks that follow the interests and experiences of the students. The learning environment must allow the students to be creative and encourage them to discuss the problems posed to develop their self-expression skills, including the ability to use the terms of natural science. Students' environmental attitudes are deepened.

III In this school stage, students learn to describe objects and phenomena quantitatively and develop skills in processing information analytically. In developing research skills, special attention is paid to planning and carrying out studies and analysing, interpreting and presenting results, including using digital tools and online environments. The perception that there is no single universal scientific method that is used to obtain new knowledge is formed. Research is carried out both with actual substances, objects and tools, as well as using computer simulations and secondary sources of information. The students learn to evaluate the reliability of different types of information sources and distinguish scientific information

from non-scientific information.

In addition to practical activities, various theoretical tasks are solved to develop students' abstract thinking. Homework consolidates the knowledge gained in lessons and leads students to apply the knowledge to everyday activities. Various problem-based learning methods are applied to form higher-level thinking and attitudes, including discussion, role play, case studies, the design of artificial objects or solutions, etc.

In both the second and third school stages, it is important to maintain students' learning motivation and to develop their interest in studying science and professions related to natural sciences, form their understanding of the importance of natural sciences and technology in everyday life as well as the need for research in society.

2.1.2. Knowledge, skills and attitudes to be acquired by the end of the school stage

First school stage	Second school stage	Third school stage
The students: 1) have an interest in nature and its study and understand the necessity of knowledge in natural sciences; 2) formulate their cognitive experiences, describe phenomena and properties of objects, use the natural science terms they have learned in speech and written texts; 3) make easier observations, perform practical work, observe safety requirements; formulate observation information, draw conclusions, and present the results under the guidance of a teacher; 4) notice and formulate problems in their immediate environment and propose solutions, make reasoned decisions using creative and critical thinking; 5) find information about natural sciences under the guidance of a teacher, use media and technology tools for data collection, learning and co-operation; 6) understand that scientific knowledge is obtained through observations and experiments, know professions related to the natural sciences; 7) behave safely and lead a healthy lifestyle, value spending time in nature and the biodiversity of their home area, notice the beauty and uniqueness of nature and treat it with respect, care for living beings and their needs.	The students: 1) are interested in nature, its study and the study of natural sciences; 2) observe and describe natural and man-made objects and explain natural phenomena using terms, symbols and units related to natural sciences; understand simpler natural scientific texts; 3) use or build a model to demonstrate an understanding of processes and systems; 3) plan simple practical work under the guidance of the teacher, pose research questions and check hypotheses, follow safety requirements and select appropriate measuring instruments; analyse data, draw conclusions and present research results; 4) notice and formulate problems related to their everyday life and propose solutions, make reasoned decisions using creative and critical thinking; 5) find information on natural sciences and technology; evaluate the reliability of the sources with the help of a teacher; use media and technology tools for learning, data collection and analysis and collaboration; 6) understand that scientific knowledge is evidence-based and obtained as a result of systematic research.	The students: 1) are interested in the environment, its study and the study of natural sciences; 2) observe and describe natural and man-made objects and explain and justify natural phenomena; understand natural science texts, use terms, symbols and units related to natural science to explain phenomena and processes; use or build a model to demonstrate an understanding of processes and systems; 3) independently formulate and pose research problems, questions and hypotheses, design and organise a study, follow safety requirements and draw valid conclusions based on the study; present research results; 4) notice and formulate problems related to everyday life on a personal, local and global level and propose solutions, make reasoned decisions using creative and critical thinking, taking into account different aspects (scientific, social, economic, ethical); 5) find information on natural sciences and technology; critically evaluate the reliability of the sources; use media and technology tools for learning, data collection and analysis and collaboration; 6) understand that scientific knowledge is evidence-based but changes over time; understand the importance and limitations of science and natural science models; understand how science, technology and society interact; distinguish between scientific and non-scientific information and explain their differences;

	7) recognise the importance of science and technology and the risks associated with their development; 8) understand the necessity of natural scientific knowledge in everyday life and its relevance to future career choices, are familiar with professions related to the natural sciences and technology in their surrounding area; 9) understand the links between human activities and the environment in Estonia and their home area and show care and respect for all living things; value biodiversity and sustainable development; address environmental problems through citizens' initiative; behave safely and lead a healthy lifestyle	7) are motivated for lifelong learning, know career opportunities related to natural sciences and technology; 8) understand the links between human activities and the environment and show care and respect for all living things; value biodiversity and sustainable development; address environmental problems through citizens' initiative; recognise their rights and responsibilities and limits in dealing with environmental issues; behave safely and lead a healthy lifestyle.
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2.1.3. Learning Outcomes

First school stage

3. After completing the 3rd grade, students:

- 1) distinguish between objects and phenomena of living and non-living nature, as well as natural and artificial substances (materials); describe and group them based on various characteristics, relying on observations and experiments carried out;
- 2) weigh bodies, measure temperature and length, choosing the appropriate measurement instruments;
- 3) make assumptions about the properties of familiar materials and the behaviour of bodies, conduct experiments under the guidance of a teacher to verify these assumptions, and draw simple conclusions based on experiments; relate the acquired knowledge to situations occurring in everyday life;
- 4) make weather observations, formulate data and draw conclusions based on them; characterise the weather and select appropriate clothing for the weather;
- 5) notice and observe seasonal changes in nature and give examples of their importance in human life;
- 6) prepare and present a research report on a species of plant, fungus or animal;
- 7) understand that scientists ask questions and make observations and experiments to answer them, know that repeating experiments increases the reliability of the results, find the influencing factor in an experiment;
- 8) give examples of the importance of living organisms in nature;
- 9) describe the external structure, nutrition, growth and mobility of plants, animals (including

humans) and fungi and link them to the living environment;

10) distinguish organisms belonging to one species;

11) differentiate between vertebrates (fish, amphibians, reptiles, birds and mammals) and invertebrates (insects and spiders);

12) know the best-known animals, plants and fungi of their home area; describe the lifestyles and habitats of the animals studied; know how to avoid dangers associated with animals and poisonous plants and fungi;

13) give examples of the relationships between organisms in nature and compile simple food chains;

14) understand that humans are a part of nature and depend on it; give examples of how people's actions affect nature;

15) compare people's lives in the countryside and in the city;

16) understand a map; find familiar objects on the map of the school area;

17) find their home area, major uplands, islands, peninsulas, bays, rivers, lakes and cities on the map of Estonia;

18) determine directions with a compass;

19) notice the biodiversity of their home area and the diversity of the landscape and explain their importance;

20) move safely in nature, without harming nature, others or themselves;

21) consider the needs of living beings (including other people);

22) consume responsibly, avoid harm to their own health and that of others;

23) behave safely in traffic;

24) make proposals for environmental protection in their surrounding area and participate in activities in this field.

Second school stage

6. After completing the 6th grade, students:

1) prepare natural scientific models, explain objects and phenomena with the support of models: the location of continents and oceans, the structure of the solar system and the movement of the planets, the alternation of night and day and seasons, the water cycle;

2) find information about objects and phenomena, assess the reliability of information with the help of a teacher, prepare and present reviews on a selected topic (outer space, natural disasters, a lake/river in their home area, protected species/object, different regions of the world, etc);

3) formulate a natural scientific research question or hypothesis with their peers, plan and carry out research based on a body of water, settlement, forest, meadow, etc, in their home area, collect and formulate data and present research results;

4) use technical aids for research and data collection (eg microscope, digital sensors, magnifying glass, tape measure); use safe working practices in conducting experiments;

5) offer ways to increase the reliability of test results, find the influencing factor in a test;

6) discuss the need to study nature and outer space; give examples of ways scientists collect evidence;

7) use determiners to learn about species;

8) plan and conduct water purification experiments with their peers;

9) measure the mass of a substance and the volume of a liquid and prepare a solution;

10) measure air temperature, assess cloudiness and wind speed and determine types of clouds and direction of wind;

11) compare the weather in different parts of Estonia based on a weather map and characterise the air temperature, precipitation and wind direction based on the drawings;

12) locate continents and oceans, major European countries, Estonian county centres, major cities, rivers, lakes, bogs, nature reserves, bays, straits, peninsulas and islands on a map and

describe their position;

13) compare the forms of life of plants, animals, fungi and bacteria; explain their importance in nature; give examples of their effects on the human body;

14) know the most important organs in the human body, their tasks and function;

15) explain the effects of environmental conditions on living organisms (including humans); characterise the adaptations of plants and animals;

16) describe and compare the living conditions of communities (body of water, bog, forest, meadow, field/garden, settlement), know their most typical species;

17) compile food chains and food webs of communities; explain nutritional relationships in ecosystems (producers, consumers and decomposers);

18) characterise, based on experiments, the composition and properties of water, air and soil; relate them to processes occurring in nature;

19) explain how groundwater is formed and justify the need to protect it; describe options for obtaining drinking water;

20) assess the living conditions and environmental status of their home area (water, air, light, noise, waste, companion animals); make proposals to improve the environmental status of their home area; participate in activities in this field;

21) explain the operation of water, sewerage or energy systems in their home area;

22) plan a green or recreation area for their home area, draw up a plan for converting an exhausted quarry into a usable environment, plan a future settlement, etc;

23) compare the properties of materials used in everyday life and relate them to fields of use;

24) describe the relationships between force, movement and energy based on experiments;

know the types and sources of energy; evaluate the possibilities of generating and using renewable energy in their home area;

25) compare, based on experiments, the propagation of sound in different materials; correlate the pitch with the frequency of oscillation;

26) analyse their family's water or energy consumption and the generation of household waste and assess their impact on the environment; make suggestions for saving water, energy and materials;

27) give reasons for the need for the sorting and treatment of household waste, sort household waste;

28) explain the formation and development of bogs and give reasons for the abundance of bogs in Estonia;

29) describe and compare different soil samples and name components of soil; explain soil formation and its importance in nature;

30) give examples of factors affecting crop yields, causes and consequences of soil damage;

31) describe soil biota and the relationships between soil organisms;

32) relate the combustion of oxygen and carbon dioxide, decay and respiration to photosynthesis; give examples of cycle of matter in nature;

33) explain the reasons for the low salinity of the Baltic Sea and the influence of the Baltic Sea on Estonian weather;

34) discuss the use of renewable and non-renewable natural resources and environmental problems in Estonia and offer ways to solve them;

35) assess the impact of human activities on communities, discuss their importance and the need to protect them;

36) relate the study of nature and conservation and management of communities to professions involved in these fields.

Third school stage

7. After completing the 7th grade, students:

1) formulate research problems and questions and hypotheses that can be investigated

- (verified) through experiment or observation, plan and organise experiments with peers, collect data, formulate the results in tables and figures; draw valid conclusions from the data, present the results (including digitally);
- 2) distinguish between an independent and a dependent variable in an experiment; understand the need to control secondary variables;
 - 3) understand the need for repeat and verification tests; analyse the reliability of the data collected and the validity of the conclusions;
 - 4) comply with safety requirements when carrying out tests and justify their necessity;
 - 5) find information about substances, bodies, phenomena and processes being studied and assess the reliability of sources with the help of a teacher; present the results of the research;
 - 6) differentiate between scientific knowledge and unscientific knowledge;
 - 7) discuss the development and importance of natural sciences and technology in everyday life and society; give examples of their interrelationships;
 - 8) measure or calculate the area of a shape, the volume of a body, the speed of movement, density;
 - 9) distinguish substances and materials based on studying their properties (colour, density, melting and boiling points, thermal conductivity) and relate the properties to their uses;
 - 10) know that substances are made up of atoms and molecules; draw up formulas of substances based on simpler molecular models;
 - 11) prepare a solution with a certain percentage, give examples of solvents, soluble substances and solutions and explain the importance of solutions in nature and everyday life;
 - 12) separate a mixture using appropriate methods;
 - 13) discuss the importance and limitations of models, select a suitable model for explaining a specific phenomenon;
 - 14) explain how solids retain their shape and hardness, liquids their fluidity and gases their volatility with the interaction of particles;
 - 15) differentiate between physical, chemical and biological phenomena and give examples of associations between them;
 - 16) relate the phenomena of heat transfer and energy conversion to natural processes and everyday life; give examples of the validity of the law of conservation of energy;
 - 17) associate changes in states of water with precipitation (rain, snow, dew, fog, frost);
 - 18) explain, on the example of photosynthesis, respiration and combustion, that chemical reactions either absorb or release energy;
 - 19) describe the associations between the living and non-living nature on the example of the carbon cycle;
 - 20) associate adaptations with physical and chemical environmental conditions;
 - 21) analyse the possible environmental impact and ecological footprint of their activities; justify the need to conserve energy;
 - 22) justify the importance of material recovery and provide opportunities for the recovery of materials;
 - 23) consider the suitability of their interests and abilities for continuing studies in natural science or technology specialities.

2.2. Biology

2.2.1. Description of the subject

Biology plays an important role in the formation of students' natural scientific worldview. Studying biology builds on knowledge, skills and attitudes acquired in natural science and integrates learning with other natural sciences, such as chemistry, physics, geography and mathematics. An important role is played by skills of solving problems and making justified decisions.

Through studying biology, students acquire competencies in natural science and technology

along with several other vital competencies. They learn to value a sustainable and responsible way of life and acquire a permanent positive attitude towards all living beings to be an active member of civil society in the future and to be able to speak out on issues of nature and environmental protection.

Through the studies, students develop the ability to solve biological problems related to everyday life and make competent decisions, which also increases their ability to cope in the natural and social environment. The knowledge, skills and attitudes acquired in biology, integrated with those learned in other subjects, form a basis for internally motivated lifelong learning.

The study of biology has several objectives: it aims to provide an overview of living nature, the diversity of organisms, their structure and function, heredity, evolution and ecology as well as the principles of environmental protection, to develop a basic understanding of the terminology used in the various branches of biology and to explore the unique characteristics of human beings and healthy lifestyles.

In doing so, the student learns to use scientific methods characteristic of biology, which are associated with obtaining the necessary information and assessing its veracity. Learning is based on the individual characteristics of students and the versatile development of their abilities. The studies develop a positive attitude towards biology as a natural science and cultural phenomenon, which is expressed, among other things, in a consciously responsible and sustainable attitude towards one's living environment and the consideration of ethical, moral and aesthetic aspects when solving problems of everyday life.

The study is student-centred, taking into account the age and individual characteristics of the student in the development of various forms of co-operation. One of the priorities of active learning is to grasp the scientific method and apply it in solving problems arising from the natural and social environment.

The students gain an overview of the main achievements, patterns, theories and future orientations of modern biology, which also helps them in their future career choices. In the course of the studies, students acquire the ability to use various sources of information, including electronic, and to assess the veracity of the information contained therein. In all this, students develop knowledge and skills that enable them to describe, explain and predict various natural phenomena.

To increase the student's internal learning motivation, a variety of active learning methods, forms and techniques are applied: problem-based learning and project study, role games, discussions, disputes, brainstorming, concept mapping, outdoor learning, field trips, excursions, etc. The preparation of reports and oral and poster presentations are also important. Modern information technology tools are used at all stages of study.

Practical work, including research, holds an important place in the acquisition of biological knowledge, by which students acquire the skills of posing problems, formulating hypotheses, planning experiments or observations and organising them. The latter is associated with the correct use of tools and the choice of expedient research and observation methodology. The ability to analyse the results obtained and present them in a written and oral summary is important.

2.2.2. Knowledge, skills and attitudes to be acquired by the end of the school stage

After completing the basic school, students:

- 1) explain the fundamental processes of living nature, relations between organisms and their connections to the non-living environment and use correct biology-related vocabulary;
- 2) take a responsible attitude towards the living environment, valuing biodiversity, sustainable and responsible lifestyles and the principles of sustainable development;
- 3) use knowledge of biology and the natural scientific method to solve the problems of everyday life and make reasoned decisions;
- 4) are able to formulate research questions, plan and organise observations and experiments

with safety requirements in mind, draw correct conclusions and present the results obtained orally and in writing;

5) use different sources of biological information, critically evaluate the information contained therein, distinguish it from non-scientific points of view and use scientific information to solve problems;

6) value the natural environment as part of culture, take an interest in biology and other natural sciences, understand the role of creativity and innovation in the development of science and technology, their interrelationships, limitations, risks and importance in everyday life;

7) have acquired an overview of professions connected to biology, use biology-related knowledge and skills in making career choices, and are motivated for lifelong learning.

2.2.3. Learning Outcomes

Biology as a Field of Study

The students:

1) analyse the necessity of biology-related knowledge and skills in everyday life and different professions;

2) compare animals, plants, fungi, protozoa and bacteria;

3) give examples of forms of life of different groups of organisms.

Characteristics of Vertebrates

The students:

1) associate the adaptations of mammals, birds, reptiles, amphibians and fish with their living environment;

2) analyse the importance of the adaptations of the different senses of mammals, birds, reptiles, amphibians and fish, depending on their habitat and lifestyle;

3) explain and give examples of the importance of vertebrates in nature and human activities and justify restrictions on their protection, give examples of protected species and explain the reasons for their endangerment.

Metabolism and Energy Exchange of Vertebrates

The students:

1) explain the interrelationships between metabolism and energy exchange;

2) relate different food sources of vertebrates to the ways of obtaining food and the peculiarities of the digestive system;

3) relate the particularities of the respiratory and circulatory systems of different vertebrate animal groups to being warm-blooded or cold-blooded;

4) give examples of ways of surviving adverse living conditions in warm-blooded and cold-blooded animals.

Reproduction and Development of Vertebrates

The students:

1) analyse the advantages of internal and external fertilisation and foetal development in vertebrates;

2) compare and give examples of direct development and metamorphosis;

3) relate the need to care for the offspring of vertebrates with the peculiarities of reproduction and development of different groups.

Evolution of Vertebrates

The students:

1) explain the improvement of vertebrates in the course of evolution;

2) give examples of evidence for the ancestry of vertebrates.

Features and Life Processes of Plants

The students:

1) differentiate between animal and plant cells and their main parts in drawings and analyse the functions of these parts;

- 2) analyse the consistency between the structure and function of the organs of flowering plants, correlate it with the movement of substances within the plant, the place of growth of the plant and its means of reproduction and distribution;
- 3) prepare and analyse diagrams of the precursors, end products and conditions affecting the process of photosynthesis;
- 4) explain the importance of photosynthesis and respiration in the vital functions of plants and other organisms;
- 5) compare the external structure characteristic of different plant groups and give examples of common plants in Estonia;
- 6) analyse the advantages of sexual and asexual reproduction on the example of different plants, compare different means of reproduction, pollination and distribution and give examples of these;
- 7) analyse the role of plants in ensuring the sustainability of nature as a complete system and in human activities and give examples of this.

Characteristics and Life Processes of Fungi

The students:

- 1) compare fungi with plants and animals;
- 2) describe and give examples of the diversity of the structure and function of different fungal groups and lichens, including an explanation of the importance of parasitism and symbiosis;
- 3) explain the means of reproduction of fungi and lichen and the conditions necessary for their development;
- 4) analyse the role of fungi and lichen in nature and human activities and give examples of this, valuing them as essential parts of living nature.
- 5) know the most important edible and poisonous species of fungi and recognise them in nature.

Features and Life Processes of Invertebrates

The students:

- 1) compare the structure of invertebrates and vertebrates and the characteristics of the most important groups of invertebrates, give examples of the corresponding groups of animals;
- 2) relate the external structure and adaptability of different invertebrate animals to move, breathe, feed and navigate their habitat;
- 3) analyse the benefits of unisexualism and hermaphroditism in different groups of invertebrates and explain and give examples of complete or partial metamorphosis;
- 4) explain the importance of changing the host organism, food source and living area during the development of parasitic organisms and give examples;
- 5) analyse the role of different invertebrates in nature and human activities, valuing invertebrates as an important part of living nature, and give examples of this;

Evolution of Living Nature

The students:

- 1) explain the nature of biological evolution and give examples of evidence of evolution in the animal and plant kingdom;
- 2) justify the reasons for the emergence of the struggle for existence and relate the struggle for existence to natural selection;
- 3) explain the emergence of species and the basic directions of evolution of larger groups of organisms;
- 4) give examples of the most important stages of human evolution.

Ecology and Environmental Protection

The students:

- 1) explain the structure of ecosystems and biospheres and give examples of these;
- 2) analyse and give examples of the impact of factors of living and non-living nature on different groups of organisms;
- 3) analyse the effect of environmental factors on the abundance of organisms based on information given in diagrams and tables;

- 4) analyse the relationships between organisms in an ecosystem, understand the processes occurring in living nature and evaluate the positive and negative impacts of human activities on populations and the survival of ecosystems;
- 5) understand the need for green transition and notice environmental problems, find ways to mitigate them in an age-appropriate manner;
- 6) explain and value biodiversity and solve dilemmas related to the conservation of biodiversity.

Structure and Living Processes of Microorganisms

The students:

- 1) explain the peculiarities of the basic characteristics of bacteria, protozoa and viruses in comparison with plants and animals;
- 2) give examples of the distribution of bacteria and protozoa in different habitats, including aerobic and anaerobic environments; assess the importance of rapid reproduction and the formation of resting spores in the spread of bacteria;
- 3) analyse and explain the importance of bacteria and protozoa in nature and human activities;
- 4) explain how to protect food from bacterial spoilage;
- 5) associate the most common human diseases caused by bacteria, viruses and protozoa with their modes of transmission and know how to avoid them.

Human Tissues and Organs

The students:

- 1) compare and give reasons for the relationship between the structure and functions of different tissues; give examples of tissues and organ systems of different organs;
- 2) analyse the accordance of the structure and function of skin when fulfilling tactile, protective, thermoregulatory and excreting functions; value a healthy lifestyle related to skin health.

Bones and Muscles

The students:

- 1) differentiate, in a drawing or model, the main bones and muscles of humans;
- 2) explain the relationship between the structure and function of bones and muscles, compare the structure and function of smooth muscle, striated muscle and cardiac muscle;
- 3) analyse the relationships of various ligaments with their functions and give examples of them;
- 4) analyse and give examples of the impact of proper nutrition and exercise on the musculoskeletal system; attach importance to healthy exercise.

Circulatory System

The students:

- 1) analyse drawings and diagrams of the human circulatory system;
- 2) associate the structural specificities of different blood vessels and components of blood with their functions;
- 3) associate the most common cardiovascular diseases in humans with the causes of their occurrence and value a lifestyle that strengthens the circulatory system and the immune system;
- 4) explain the role of blood in the formation of short- and long-term immunity of the body, the emergence of immune disorders and the importance of vaccination to prevent infectious diseases.

Digestion and Excretion

The students:

- 1) compile and analyse drawings and schemes of the structure of the digestive system and explain, on the basis of these, the digestion of food and absorption of nutrients;
- 2) explain the tasks of proteins, fats, carbohydrates, vitamins, mineral products and water in the human organism and the problems caused by over- or under-consumption;
- 3) assess the role of the kidneys, lungs and skin in the excretion of waste products.

Breathing

The students:

- 1) analyse the accordance between the structure and function of the respiratory system;
- 2) compile and analyse drawings and schemes of the structure and function of the respiratory system and the composition of the air inhaled and exhaled;
- 3) explain the nature of breathing, including the role of oxygen in cells, inhalation and exhalation, and the regulation of breathing;
- 4) analyse the effect of exercise on the respiratory system;
- 5) explain the main causes of the most common illnesses of the respiratory organs and ways of preventing these illnesses.

Reproduction and Development

The students:

- 1) compare the structure and function of the male and female sex organs;
- 2) compare the structure and development of human ova and sperm cells, explain the fertilisation of the ovum and the factors affecting it, and give examples of changes in foetal development;
- 3) associate age-specific anatomical changes in the body with functional changes.

Regulation of Functions

The students:

- 1) explain the structure and functions of the central nervous system and peripheral nervous system;
- 2) relate the structure of the nerve cell to its function, compile and analyse the schemes of the reflex arc and, based on this, explain its function;
- 3) associate different endocrine glands with the function of hormones they produce;
- 4) explain the role of the nervous system and hormones in the regulation of the functions of organ systems;
- 5) take a critical attitude towards the abuse of substances that damage the nervous system.

Exchange of Information with the Outer Environment

The students:

- 1) analyse the co-operation between the parts of the eye and the centre for vision of the cerebrum in the creation and interpretation of vision;
- 2) explain the reasons for the occurrence of short- and long-sightedness and ways of preventing and correcting visual disorders;
- 3) associate the structure of the ear with the sense of hearing and balance, and value a lifestyle that spares the sense organs;
- 4) compare and associate the structure and function of the organs connected to the senses of smell and taste.

Heredity

The students:

- 1) analyse the role of heredity and changeability in the example of human features;
- 2) explain the connection between DNA, genes and chromosomes and their role in heredity and the inheritance and expression of genes;
- 3) solve simple genetics tasks related to the expression of dominant and recessive gene alleles;
- 4) assess the role of hereditary and non-hereditary changeability in the example of human features and analyse information about the extent of changeability presented in diagrams and tables;
- 5) give examples of the fields of activity of genetic engineering and assess the possibilities of genetic modification of organisms based on scientific and other valid points of view;
- 6) give examples of the possibilities of prevention of hereditary diseases and diseases with hereditary predispositions and analyse them;
- 7) explain the hereditary and non-hereditary diversity of people and have a sympathetic

attitude towards it.

2.3. Physics

2.3.1. Description of the subject

Physics is a natural science and plays an important role in the shaping of students' natural scientific and technology literacy. Physics explains natural phenomena, creates relevant models and is closely related to mathematics. Physics forms a basis for understanding instruments and technology and helps students to value technological professions. The learning of physics in basic school only covers a small proportion of physical phenomena and forms a basis from which a complete picture of physics as a natural science is later developed. In studying physics, students obtain a preliminary overview of the language of physics and learn to use it.

The things studied in physics are associated with everyday life, mathematical skills, instruments and technology and other natural sciences. In studying physics, the integration of natural sciences (physics, chemistry, biology and geography) is based on two directions. Vertically, these subjects are integrated through common topics, such as development (evolution), interaction, movement (changing and transformation), systems and structures, energy, technology and the environment (society). Vertical integration is supported by the specificity of the area, taking into account the horizontal integration of the subjects. The value judgements of students are shaped by associating the solutions to problems with the general cultural/historic context. At the same time, the role of physicists in the history of science is studied as well as the meaning of physics and its applications for the development of humankind.

Solving calculus-, graphics- or problem-based tasks and assessing how realistic the results are creates a basis for critical thinking. When learning about phenomena, experiments are preferred; when solving problems, scientific methods are used. The material studied is presented in a problem-based way as much as possible and in connection with the students' everyday lives. In the studies, the individual specificities of the students are taken into account and their abilities are developed from diverse perspectives, great attention is paid to the development of students' learning motivation. To this end, different active learning methods are used: problem- and research-based learning, project study, discussions, brainstorming, role play, outdoor learning, field trips, etc.

When planning studies, the teacher may change the order of the topics studied, wherein it is important to note that the changed order of topics must follow the developmental specificities of the students and that the teaching takes place according to the principle of the growth of abstraction. At all stages of study, technological tools and ICT opportunities are used. With research-based study the students learn how to formulate problems and hypotheses, plan work, carry out observations, measure and process, interpret and present results. An important role is played by oral and written presentation of research results, including verbal and visual forms of presentation. An important role is also played by the skill of using different sources of information (including the internet), critically assessing the information found therein, and referring to the sources used.

2.3.2. Knowledge, skills and attitudes to be acquired by the end of the school stage

In addition to the general learning outcomes described in the competencies in the related field, the study of physics in basic school results in the students:

- 1) understanding important models in physics;
- 2) applying formulas for the quantitative description of physical phenomena and the properties of bodies;
- 3) preparing graphs, drawings and diagrams for describing physical phenomena and analysing

information presented in diagrams;

- 4) explaining and giving reasons for physical phenomena and the properties of bodies based on models of physics;
- 5) using different sources to find information related to physics and assessing the reliability of the sources;
- 6) safely designing and conducting experiments to study physical phenomena and the properties of bodies, analyse test results and draw reasoned conclusions.

2.3.3. Learning Outcomes

Study of Light

The students:

- 1) recognise different light sources; classify light sources according to their size and spectral composition of light;
- 2) know the law of rectilinear propagation and reflection of light and construct drawings explaining optical phenomena and can organise relevant experiments based on this knowledge;
- 3) apply the principle of light refraction to explain the principle of operation of a lens and solve problem-solving tasks;
- 4) explain the concept of focus, focal length and optical strength;
- 5) know the properties of various lenses and associate the formation of images with the properties of lenses; construct the course of rays in a convex and concave lens, distinguish between real and virtual images;
- 6) relate the spectrum of reflected light to the colour of objects;
- 7) explain the principle of operation of various optical devices based on drawings;
- 8) explain the principle of operation of the eye as an optical system and the causes of near-sightedness and far-sightedness;
- 9) apply the following relationship when solving problem-solving tasks $D = \frac{1}{f}$.

Mechanics

The students:

- 1) study and describe the movement of the body and can analyse it on a diagram;
- 2) study and describe the interaction between bodies and explain the change in the speed of bodies depending on the mass of the bodies and the duration of the interaction;
- 3) know that the strength of interaction is characterised by force;
- 4) compare the force of gravity acting on different bodies and relate it to the mass of the body;
- 5) study friction force and explain its effect on the movement of bodies, analyse the dependence of friction force on normal force on a diagram;
- 6) study elastic force and explain the reasons for its occurrence;
- 7) use a dynamometer to measure different forces;
- 8) design and carry out an experiment to determine the pressure and relate the pressure to the surface area and applied force;
- 9) describe the transmission of pressure in gases and liquids (Pascal's law); conduct an experiment to study buoyancy acting on bodies in liquids and explain the results of the experiment;
- 10) know the conditions of floating and sinking of bodies and explain their relationship with natural phenomena;
- 11) explain the concepts of air pressure, pressure head and buoyancy and apply them when explaining natural phenomena;
- 12) explain the concept of mechanical work, mechanical energy (potential and kinetic energy), power and efficiency;
- 13) explain the purpose and general principle of operation of simple mechanisms, applying the golden rule of mechanics;
- 14) describe oscillation based on a model, using the concepts of amplitude, period and

frequency;

15) associate oscillation with the formation of sound and the propagation of sound waves;

16) design and organise experiments to measure the volume of noise and study the relationship between pitch and frequency of the sound of musical instruments;

17) apply the following relationships when solving problem-solving tasks: $v = \frac{s}{t}$; $\rho = \frac{m}{v}$; $F = mg$; $p = \frac{F}{S}$; $p = \rho gh$; $F_{\text{buoy}} = \rho gV$; $A = Fs$; $N = \frac{A}{t}$; $f = \frac{1}{T}$.

Electricity

The students:

1) explain electrification and electrical interaction of bodies;

2) know the concepts of electric charge, elementary charge, electric charge of the body, electric field, electric current, free carriers, electric conductor and insulator and apply these to explain natural phenomena;

3) study and describe electric current in aqueous solutions of electrolytes and in metals;

4) name the parts of the circuit and explain their purpose; draw up simpler electrical diagrams;

5) explain the purpose and rules of use of electrical appliances and electrical measuring equipment (ohmmeter, ammeter, voltmeter, electricity meter);

6) design and carry out experiments to measure current, voltage, electrical resistance and resistivity;

7) examine the relationships between voltage, amperage and resistance values of the circuit components in the case of series and parallel connection and analyse the results obtained;

8) design and conduct experiments to calculate the operation and power of electric currents and analyse the results obtained;

9) determine the total power of current-using equipment, assess its compliance with the installed fuse and calculate the value and cost of the energy consumed;

10) explain the concepts of short circuit, fuse and protective earth;

11) describe the properties of magnets and the magnetic field and relate them to the magnetic field of the Earth and other magnetic phenomena;

12) relate electric current and magnetic phenomena using examples and applications in technology;

13) apply the following relationships when solving problem-solving tasks: $I = \frac{U}{R}$; $I = I =$

$$I; U = U_1 + U_2; R = R_1 + R_2; I = I_1 + I_1; U = U_1 = U_2; \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}; R = \frac{\rho l}{S}; A =$$

$$IUt; N = IU; Q = I^2 R t$$

Thermodynamics and Nuclear Energy

The students:

1) relate the temperature of the body and the thermal expansion of bodies to the heat motion of matter particles;

2) explain the purpose and rules of use of the thermometer, as well as various temperature scales;

3) distinguish the types of heat transfer in explaining natural phenomena: thermal conductivity, convection and thermal radiation;

4) explain the change in internal energy during the heating and cooling of bodies;

5) explain the concepts of heat and specific heat capacity of a substance and design an experiment to determine the specific heat capacity of a body;

6) analyse everyday life and natural phenomena according to the thermal properties of bodies and the main characteristics of heat transfer;

7) explain the change in the internal energy of the body during fusion, solidification, evaporation and condensation;

- 8) explain the meaning of heat of fusion, heat of boiling and calorific value;
- 9) solve and analyse complex tasks in thermal physics with applicable content that can be reduced to partial tasks;
- 10) relate the composition of isotopes, radioactive decay and nuclear reaction to the construction of the atomic nucleus;
- 11) explain the practical value of nuclear fusion and fission;
- 12) characterise and compare α -, β - and γ -radiation;
- 13) identify the sources of natural ionising radiation and explain the dangers associated with it;
- 14) apply the following relationships when solving problem-solving tasks: $Q = cm(t_2 - t_1)$;
 $Q = \lambda m$; $Q = Lm$.

2.4. Geography

2.4.1. Description of the subject

Geography is essential in the development of students' natural scientific literacy and all general competencies. Learning builds on knowledge, skills and attitudes previously acquired in natural science. Geography creates good prerequisites for cross-disciplinary learning as well as for the integration of natural and social sciences, helping students to see the connections between knowledge gained in mathematics, physics, biology and chemistry as well as in history and citizenship education.

By learning geography, students get an overview of the phenomena and processes taking place in nature and society, their spatial distribution and interconnections. Students will develop an understanding of the Earth as a whole and the interconnections between the environment and human activities on a personal, local and global level. Familiarity with different regions of the world allows students to understand the uniqueness of each place and at the same time the global connectedness of places, which means that the same processes can function differently in different places depending on the natural, economic or social circumstances. Students' spatial thinking and spatial analysis skills develop as they study geography.

In geography lessons, students can discuss relevant and important societal topics that help them make sense of their knowledge of the subject. This creates the prerequisites for the formation of active and conscious members of society who notice the problems of everyday life and can offer reasoned solutions to them. The studies develop the ability to assess the direct and indirect consequences of one's decisions or actions.

In the information society, skills in the use of information technology and critical thinking are increasingly important. In geography lessons, students learn to apply a variety of sources of information, including map applications and data portals, and critically evaluate the reliability of information.

The studies are based on exploratory learning which develops students' problem-solving and research skills. They learn to notice problems, formulate hypotheses and research questions, plan and organise research, as well as collect data through observations, surveys or interviews, but also from secondary sources: maps, satellite photos, data portals, etc. Students develop analysis, generalisation and inference skills by processing data. Written and oral expression skills, including compiling correct natural scientific text and using subject-specific vocabulary, are developed through the interpretation and presentation of research results.

By studying geography, students begin to understand the essence and importance of the science of geography in everyday life and in the development of society. The students learn to notice spatial relationships and understand the possibilities of modern technology by observing both natural and social processes, modelling and creating scenarios for the future.

Geography contributes to the formation of students' values and attitudes. Approaches to world nature, population and cultural geography form a basis for an understanding and tolerant attitude towards the cultures and traditions of other countries and peoples. Studying Estonian geography forms a basis for valuing the nature, history and cultural heritage of the students'

homeland. Studying both natural and social geography develops students' environmental awareness, emphasises the importance of biodiversity, cultural diversity and sustainable economy, and values a sustainable and responsible lifestyle.

In order to develop and maintain the intrinsic learning motivation of students, teaching materials are presented in as problem-based a manner as possible and associated with the student's everyday life and home area. Fieldwork is important in geography, enabling the possibility to examine local conditions and problems, involve students in community projects and teach students to associate theoretical knowledge with practical skills. The study is based on the personal characteristics of students and the versatile development of their abilities. Diverse learning methods are applied: project study, discussions, brainstorming, role play, outdoor learning, field trips, visiting museums, etc. At all stages of study, modern media and information technology tools are used.

Geography helps us to value the many professions that require knowledge of both nature and society and the ability to work with spatial data and see interrelationships.

2.4.2. Knowledge, skills and attitudes to be acquired by the end of the school stage

After completing the basic school, students:

- 1) are interested in geography and other natural and social subjects, are motivated to study them;
- 2) use the knowledge and skills acquired in geography to explain and analyse phenomena occurring in nature and society, their spatial location and interrelationships;
- 3) notice and solve geographical problems related to their everyday life, make reasoned decisions using creative and critical thinking;
- 4) design and organise studies, formulate research questions, process and formulate data, draw conclusions and present results;
- 5) find geographical information in information sources, assess its reliability, use media and technological tools in learning and co-operation;
- 6) understand the nature and importance of the science of geography in everyday life and in the development of society;
- 7) value natural and cultural diversity and a sustainable living environment, behave safely and adhere to the principles of sustainable development;
- 8) have gained an understanding of geography-related professions and career opportunities and are motivated for lifelong learning.
- 9)

2.4.3. Learning Outcomes

Use of Maps

The students:

- 1) use both paper and digital maps and other spatial information models to find information, characterise objects and phenomena, analyse, draw conclusions and make spatial decisions and justify them;
- 2) can read a map: understand the legend and processes depicted on a map, measure distances, determine directions, geographical coordinates, time differences, etc;
- 3) navigate on a map: find countries, capitals, tectonically active regions, major landforms, bodies of water, climate zones, geographical zones, etc;
- 4) navigate and move around in a landscape using a map;
- 5) prepare a map or another model that gives spatial information.

Geology

The students:

- 1) characterise, based on drawings or a map, the internal structure of the Earth and the structure of the Earth's crust, the movement of plates and the geological processes occurring in the edge areas of the plates;
- 2) know the causes, consequences and accompanying phenomena of earthquakes and

volcanism, as well as their impact on the environment, know how to behave in the case of possible danger;

- 3) characterise and compare sediments and rocks of different formation, know the possibilities of their use;
- 4) know the importance of weathering in nature, associate weathering with the properties of rocks and the climate;
- 5) relate the location and formation of rocks and sediments, including mineral resources, to the geological structure of Estonia;
- 6) relate the formation of soils to the conditions of their formation using examples from Estonia.

Relief

The students:

- 1) compare the landforms and relief in their home area, Estonia and the world using maps and other information sources;
- 2) explain the formation and change of landforms and relief as a result of various factors, including human activities;
- 3) analyse the interrelationships between relief and human activities and consider the relief and resulting hazards when moving on a landscape.

Climate

The students:

- 1) describe the weather using a weather map and explain the movement of air and the formation of precipitation depending on the characteristics of the air;
- 2) explain the differences in climate depending on the distribution of solar radiation on Earth, general air circulation and the influence of the oceans, including currents and relief;
- 3) characterise a climate graph and relate it to the corresponding climate zone;
- 4) compare the climate of different places using thematic maps and climate graphs, relate it to the influence of factors shaping the climate and to the possibilities of human activity;
- 5) understand the impact of human activities, including land use, on the climate at both local and global levels;
- 6) know the potential consequences of climate change and the possibilities of adaptation to climate change.

Bodies of Water

The students:

- 1) understand the interrelationships between bodies of water and human activities, the importance of analysing bodies of water and the need to protect water;
- 2) analyse the water cycle in different regions of the Earth, correlate it with climate, water availability and the possibilities of human activity;
- 3) compare seas, including the Baltic Sea, rivers or lakes using information sources, and give reasons for their differences and similarities;
- 4) relate the water-consuming, transporting and accumulating activities at different sections of a river to the relief and flow rate;
- 5) relate changes in river water levels, including floods, and their extent to the climate and relief of the region;
- 6) characterise problems related to the formation and use of groundwater in their home area or Estonia, using information sources.

Geographical Zones

The students:

- 1) characterise and compare the natural components of geographical zones (ice belt, tundra, temperate zone coniferous and mixed forests, temperate zone grasslands, subtropical dry forests, deserts, savannas, rainforests) and the relationships between them, using information sources;
- 2) analyse the interaction between nature and human activities in geographical zones and the associated environmental problems.

Population

The students:

- 1) analyse the population and population processes of their home area, Estonia or a European country based on data obtained from data portals;
- 2) analyse the gender and age composition of the population of a certain region according to the population pyramid and its impact on society;
- 3) know the directions of migration in Estonia and Europe and their causes, analyse the impact of migration on society;
- 4) discuss Estonian population policy measures.

Settlement

The students:

- 1) characterise and compare trends and stages of urbanisation in Estonia and Europe and the consequences of urban growth and decline;
- 2) analyse the location and density of population in their home area, Estonia and Europe using a map, relating it to the impact of natural and social factors;
- 3) analyse the development of Estonian settlements, the living environment and the natural and socio-economic factors affecting it using information sources and offer solutions for improving the living environment of a settlement.

Economy

The students:

- 1) understand the nature and importance of sustainable economy, provide examples of sustainable management, including circular economy;
- 2) analyse the impact of natural resources, labour, capital, markets and supply chains on the Estonian and European economies;
- 3) analyse changes in the structure of the Estonian economy and relate it to the general trends of economic development;
- 4) characterise the impact of globalisation and multinational companies on the Estonian economy;
- 5) discuss problems related to economic activity based on economic, social and environmental aspects.

Agriculture and Food Production

The students:

- 1) understand the connections and importance of sustainable agriculture and food production;
- 2) characterise a food production chain, know the advantages of local food products and value Estonian products;
- 3) characterise the growing conditions, cultivation and use of certain cultivated plants, using information sources;
- 4) compare production in different plant cultivation and livestock farms and the impact of small- and large-scale production on the environment, including changes in landscapes;
- 5) characterise the prerequisites for the development of agriculture in Estonia and explain the structure of agriculture and food production.

Forestry and Forest Management

The students:

- 1) know the importance of forests and sustainable forest management and value forests as an ecosystem;
- 2) explain the role of forest management and industry, including the valorisation of timber, in the Estonian economy.

Energy Management

The students:

- 1) analyse energy consumption at their family level and in the society, value sustainable energy consumption and offer solutions for this;
- 2) analyse the advantages and disadvantages of the use of different energy carriers, including

- their impact on the environment;
- 3) have gained an overview of energy management in their home area, Estonia and Europe, and the related problems.

Service Sector

The students:

- 1) analyse the location of jobs and the availability of services in settlements at different levels of the settlement system, including in their home area;
- 2) characterise the Estonian transport system, analyse the advantages and disadvantages of modes of transport and the impact of transport on the environment;
- 3) analyse the availability of public transport in a certain settlement based on information sources and its impact on people's daily lives;
- 4) characterise and analyse the possibilities for the development of tourism in their home area, Estonia or a European country, the tourism industry and its impact on economic and social life and the environment, using information sources.

2.5. Chemistry

2.5.1. Description of the subject

Chemistry plays an essential role in the formation of students' natural scientific worldview. Learning chemistry builds on knowledge, skills and attitudes previously acquired in natural science. It is important to relate the learning to other natural sciences (mainly physics and biology) and mathematics. In studying chemistry, students acquire a simple but complete understanding of the chemical processes that take place in nature and artificial environments as well as those used in human activities, their basis, interrelationships and impact on the living environment. Learning develops the skills to solve everyday problems and make competent decisions; these skills allow students to cope in the natural and social environment. Students' engagement with the surrounding environment is important for effective learning. The knowledge, skills and attitudes acquired in chemistry, together and integrated with those learned in other subjects, form a basis for internally motivated lifelong learning.

In the course of studying chemistry, students develop a general idea of particles of matter, the diversity of substances, the composition, properties and transformation of substances, and the possibilities of applying substances and their transformations. This provides an overview of the modern technology- and energy-related problems and future directions in chemistry, which help the students in their future choice of profession. Learning about substances and their transformations helps to understand the role of scientific and technological progress in shaping the living environment and, at the same time, guides students to contemplate the importance of the responsible use of resources. Studying chemistry helps to understand the relationship between a clean natural environment and health, develops students' respect for nature and a sense of responsibility to maintain and protect the living environment. The studies develop the ability to assess the direct or indirect consequences of one's decisions or actions.

In studying chemistry, students use an inquiry-based approach based on the scientific method and solve problems arising from the natural, technological and social environments. Studying chemistry develops the skills of logical thinking, analysis, generalisation, understanding of causal relationships and approaching problems creatively. The students acquire an ability to understand and compile chemistry-related texts, make sense of and use chemistry-related vocabulary correctly, present chemistry-related information (including research results) in oral and written presentations using different verbal and visual forms of presentation (verbally and as diagrams, graphs, models and formulas), and use different sources of information, including electronic ones. All this provides students the autonomy necessary for meaningful learning.

In the studies, significant attention is paid to shaping and maintaining the students' internal learning motivation. Through practical work, students acquire the skills necessary for such

work: they learn to safely use experimental equipment and chemicals necessary in the laboratory and everyday life, to assess the danger of household chemicals and the materials used in everyday life and technology to human health and the state of the natural environment. Students solve calculation tasks in chemistry to better understand chemical phenomena and their corresponding quantitative relationships and develop logical thinking and mathematics skills. Solving calculation tasks guides students to make reasoned conclusions and decisions.

2.5.2. Knowledge, skills and attitudes to be acquired by the end of the school stage

After completing the basic school, students:

- 1) observe, understand and are interested in phenomena related to chemistry in everyday life, in the environment and in practical human activities;
- 2) implement the necessary safety precautions when using chemicals and materials in everyday life;
- 3) use chemistry terminology and symbols correctly; understand and prepare texts related to chemistry;
- 4) understand the information included in chemical equations and compile chemical equations;
- 5) use the periodic table of chemical elements, solubility table and reactivity series of metals and find the values of physical quantities from tables and diagrams;
- 6) plan and safely conduct chemical experiments to learn about the properties of substances and the laws of nature;
- 7) make calculations based on metabolic formulas and chemical equations, as well as the composition of solutions; assess the correspondence of the calculation results to reality.

2.5.3. Learning Outcomes

What Does Chemistry Involve?

The students:

- 1) know the options for incurring chemical reactions, recognise a chemical reaction by its characteristics;
- 2) comply with safety requirements when conducting experiments in the laboratory and when using chemicals in everyday life;
- 3) know the most important laboratory equipment and use it correctly in practical work;
- 4) differentiate between and prepare solutions and dispersions and give examples of solutions and dispersions in nature and everyday life;
- 5) solve calculation problems based on the percent composition of a solution.

Atomic Structure and Periodic Table: Composition of Substances

The students:

- 1) explain the structure of an atom and use the periodic table of chemical elements to find the properties of chemical elements, describe the structure of an atom and draw up an electron configuration diagram;
- 2) know the classification of chemical elements into metals, nonmetals and noble gases, find examples of the use of metals and nonmetals in everyday life from the internet and compare their properties;
- 3) differentiate between simple substances and chemical compounds and explain the composition of a substance based on the formula of the substance;
- 4) differentiate between ions and neutral atoms and explain the occurrence and charge of ions;
- 5) explain the differences between covalent, ionic and metallic bonds.

Oxygen and Hydrogen: Oxides.

The students:

- 1) explain the role of oxygen in combustion and living nature, analyse the importance of the ozone layer and its destruction due to pollution;
- 2) compare the main properties of oxygen and hydrogen;

- 3) collect gas by selecting the appropriate technique based on the solubility of the gas in water and the density of the gas in relation to the density of the air;
- 4) determine the degrees of oxidation of elements on the basis of the formula of a substance, prepare formulas based on the names of oxides and names based on formulas;
- 5) understand the principle of balancing chemical equations;
- 6) arrange combination reactions of simple substances with oxygen and prepare the corresponding chemical equations, give examples of the most well-known oxides in everyday life and their importance.

Acids and bases as substances of opposing properties

The students:

- 1) distinguish oxides, acids, hydroxides and salts based on the formula;
- 2) prepare the formulas of acids, hydroxides and salts on the basis of their names and vice versa;
- 3) correlate the acidic and alkaline properties of solutions with the particles present in them, evaluate the environment of a solution by an indicator and the pH value of the solution;
- 4) understand the opposite nature of acids and bases, arrange reactions between acids and bases and draw up the corresponding chemical equations;
- 5) give examples of well-known acids, bases and salts in everyday life.

Best-known Metals

The students:

- 1) distinguish between active, moderately active and low-active metals according to their position in the ranking of metals and study the activity of metals;
- 2) study the effect of factors affecting the rate of reactions between a metal and an acid;
- 3) associate redox reactions with changes in the oxidation states of chemical elements in the reaction, know the behaviour of metals in chemical reactions as a reducing agent and the behaviour of oxygen as an oxidizer;
- 4) prepare chemical equations for reactions between metals and acids;
- 5) assess the possibilities of application of iron, aluminium, copper and their alloys in everyday life, relating the uses to the characteristic properties of the respective materials.

Main Classes of Inorganic Compounds

The students:

- 1) understand and compile chemistry texts on the properties of inorganic compounds and the relationships between substances;
- 2) study the properties of solutions of strong and weak acids and explain the differences;
- 3) study the chemical properties of acidic and basic oxides: acidic oxide + water, (strongly) alkaline oxide + water, basic oxide + acid, acidic oxide + base; prepare equations for the respective reactions;
- 4) explain the impact of temperature on the solubility of gases and (most) salts in water, use the solubility graph and solubility table of substances to find the necessary information and make calculations and conclusions;
- 5) explain the occurrence of the most important inorganic compounds in nature and their use in everyday life (fertilisers, water hardness, building materials);
- 6) know the sources of chemical pollution and analyse reasons for pollution, environmental problems caused by pollution (acid precipitation, heavy metal compounds, over-fertilisation) and possible measures to save the environment.

Amount of Substance: Mole Calculations

The students:

- 1) make calculations based on the relationships between the amount of substance, mass and gas volume, correctly use the corresponding units, and logically justify the calculation steps;
- 2) analyse the qualitative and quantitative information included in the chemical equation, understand the conservation of mass in chemical reactions;
- 3) solve calculation tasks based on chemical equations, following the chemical equation coefficients (the mole ratio of substances) and the amounts of the substances involved in the

reaction (in moles), make recalculations, where necessary, on the basis of the relations between the amount and mass of substances and (gas) volume, and explain the process of calculation;

4) logically assess the validity of calculation results, draw conclusions and make decisions on the basis of the calculation results.

Carbon and Carbon Compounds

The students:

- 1) compare and explain the properties of the simple substances of carbon and compare the properties of carbon oxides;
- 2) know the reasons for the multiplicity of carbon compounds;
- 3) compile structural formulas and molecular models of carbon compounds according to the provided number of atoms (C, H, O), distinguish between linear, branched and cyclic carbon chains;
- 4) classify materials into hydrophilic and hydrophobic;
- 5) describe the forms of occurrence of hydrocarbons in nature and explain their uses;
- 6) distinguishes between hydrocarbons, alcohols and carboxylic acids based on the structural formula;
- 7) prepare chemical equations for the complete combustion of hydrocarbons and ethanol;
- 8) study the chemical properties of acetic acid;
- 9) know the physiological effect of ethanol and analyse the problems connected to it in everyday life.

The Role of Carbon Compounds in Nature and Carbon Compounds as Materials

The students:

- 1) explain and study the thermal effect of chemical reactions;
- 2) analyse the possibilities of using carbon compounds as fuel and distinguish between renewable and non-renewable sources of energy;
- 3) recognise polymers based on the structural formula;
- 4) understand the role of saccharides, fats and proteins in organisms, examine their properties and content in foodstuffs;
- 5) characterise the best-known polymeric materials based on carbon compounds (fibres, plastics), analyse their main properties, potential for use and the environmental problems associated with their use;
- 6) understand the necessity of having a sustainable attitude towards the living environment and analyse the possibilities of sustaining the environment.